



DJANGO INTERVIEW QUESTIONS WITH ANSWERS

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1. What is Django?

Django is a high-level Python web framework that follows the MTV (Model-Template-View) architecture and promotes rapid development with clean, reusable code.

2. Why is Django called a "batteries-included" framework?

Because it comes with built-in features like authentication, ORM, admin panel, session management, and security without needing third-party tools.

3. What is MTV architecture in Django?

- **Model: Handles database**
- **Template: Handles UI (HTML, CSS)**
- **View: Business logic (connects Model & Template)**

4. What is the difference between Django and Flask?

- **Django: Full-stack, batteries-included, opinionated.**
- **Flask: Lightweight, flexible, requires manual setup.**

5. What are Django Models?

Models define the structure of database tables using Python classes. Django ORM automatically maps them to the database.

6. What is Django ORM?

The Object Relational Mapper lets you interact with the database using Python objects instead of writing SQL queries.

7. What are Migrations in Django?

Migrations are Django's way of propagating changes made to models into the database schema using commands like makemigrations and migrate.

8. What is the use of settings.py file?

It contains configuration details like database settings, installed apps, middleware, templates, static files, etc.

9. What is the purpose of urls.py?

It defines URL patterns and maps them to corresponding views.

10. Difference between select_related and prefetch_related?

- **select_related:** Used for single-valued relationships (ForeignKey, OneToOne).
- **prefetch_related:** Used for multi-valued relationships (ManyToMany, reverse ForeignKey).

11. How do you create a Django project and app?

- **Create project:** `django-admin startproject projectname`
- **Create app:** `python manage.py startapp appname`

12. What is Django Admin?

A built-in admin panel that allows managing application data (CRUD) without writing extra code.

13. What are Django Middlewares?

Middlewares are hooks that process requests and responses globally (e.g., Authentication, Security, Session).

14. What are Static Files in Django?

Static files (CSS, JS, images) are stored separately and served using Django's static files framework.

15. What are Django Templates?

Templates are HTML files that dynamically render data using Django Template Language (DTL).

16. How does Django handle security?

- ✓ Built-in CSRF protection
- ✓ SQL injection protection via ORM
- ✓ XSS protection
- ✓ Clickjacking prevention

17. What is the difference between get() and filter() in Django ORM?

- **get():** Returns a single object, raises error if not found.
- **filter():** Returns a queryset (multiple objects or empty).

18. How does Django handle sessions?

Django provides session middleware that stores session data on the server and uses cookies to identify users.

19. What is the difference between @login_required and PermissionRequiredMixin?

- **@login_required:** Restricts view access to logged-in users.
- **PermissionRequiredMixin:** Restricts access based on user permissions.

20. How can you deploy a Django project?

Common deployment steps:

- **Configure WSGI/ASGI (Gunicorn, Daphne, Uvicorn)**
- **Use a web server (Nginx/Apache)**

- **Setup database, environment variables**
- **Use Docker/Cloud services for scaling**